

Abhishek Pandey

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WORK EXPERIENCE

PayPay India

Backend Engineer II

India

August 2025 – Present

- Designed a **high-throughput notification** scheduler leveraging virtual threads and batch processing to sustain 10K+ TPS, with end-to-end observability ensuring reliability for large-scale, event based campaigns, **driving monetization**.
- Optimized notification delivery by replacing templates with template IDs and removing S3 uploads, **reducing latency for push/email** channels, enabling template-level CTR tracking and **improved analytics for marketing team**.
- Implemented Redis-based deduplication using idempotency keys for timeline notifications, reducing duplicate notifications by around 2% while handling around 100-150M events/day.
- Reduced **cloud infrastructure costs** and strengthened security by migrating legacy caches to Redis (Valkey), enforcing least-privilege AWS IAM policies, and upgrading the multiple services to Spring Boot 3.5.
- Led the migration of a legacy cadence-based service to a new cadence cluster, ensuring **zero data loss** and uninterrupted execution of large campaigns using blue green deployment.
- Tech Stack:** Java, Distributed Systems, Github Actions, Kafka, PostgreSQL, Cadence, Jenkins

Samsung Research

Software Engineer

Bengaluru, Karnataka

August 2022 – July 2025

- Built a reactive API Gateway with non-blocking I/O and Redis-backed authentication, **reducing thread contention** and improving response times at 1M+ daily request scale for web and mobile APIs.
- Migrated microservices from Hystrix (thread-isolated execution) to Resilience4j, eliminating unnecessary thread creation, reducing context-switch overhead, and **improving system responsiveness by 12%**.
- Designed and implemented a reliable notification delivery pipeline using **cursor-based queries** and Redis for state tracking, supporting 100K+ users and driving adoption of **consistent design patterns** across the team.
- Secured PII data** from static key to non-deterministic encryption using an internal KMS, **ensuring security**.
- Achieved **\$800 per month in infrastructure cost savings** by spearheading AWS optimization initiatives, including EC2 right-sizing, decommissioning unused load balancers, and automated policy enforcement.
- Eliminated transaction failures for 1K+ users by resolving sync issues and optimizing fee algorithms.
- Mentored** junior engineers, providing guidance on system design, computer science fundamentals and **code reviews**.

Zuno General Insurance

Software Engineer Intern

Mumbai, Maharashtra

Jan 2022 – July 2022

- Wrote an API Wrapper to fetch details from SAP for a health policy achieving a 23% faster response time.
- Improved multi-member policy processing time by 12% through database query optimization.

IIIT Delhi

Summer Research Intern

New Delhi, India

May 2021 – July 2021

- Increased search result relevance and consistency by **40%** by implementing advanced culinary category filters.
- Built and deployed **5+ REST APIs** backed by SQLite3 to enhance system accuracy and resolve critical bugs.

SKILLS

- Languages:** Java, Kotlin, C++, Python, JavaScript
- Frameworks:** Spring Boot, Spring Webflux, Spring Cloud, Node.js, React, Hibernate
- Cloud & DevOps:** AWS (EC2, S3, RDS, IAM), Docker, Kubernetes, Kafka, CI/CD, Maven, Gradle, Git, Kafka
- Databases :** NoSQL Databases, PostgreSQL, MySQL, DynamoDB, Redis, MongoDB, Cassandra, Elasticsearch
- Core Concepts:** System Design, Data Structures & Algorithms , RESTful APIs, Design Patterns, Agile Pattern
- Generative AI & Tools:** Agentic Coding(Claude, Codex), agentic workflows, MCP, AI-assisted SDLC

PROJECTS

- Blog App:** Built a secure RESTful platform with user authentication using Express.js, Node.js, and MongoDB.
- Online File Explorer:** Developed a web-based file system interface using Node.js modules (fs, path).

EDUCATION

Birsa Institute of Technology

Bachelor of Technology in Electronics and Communication Engineering | CGPA: 8.65/10

Dhanbad, Jharkhand

August 2018 – June 2022

ACHIEVEMENTS

- Recognized as a Top Performer for two consecutive quarters at Samsung India for exceptional contributions.
- Solved 800+ problems on coding platforms including LeetCode, CodeChef, and Codeforces.